

PUBLIC-PROJECT MECHANISM DESIGN • CERTIFICATE-FIRST RESEARCH

When the “fair” funding rule isn’t

An exact map of how small groups can break the textbook rule for funding shared projects — and by exactly how much.

Mohit Mendiratta • Independent research • SSRN Abstract ID 7293498

github.com/jammysunshine/research-showcase/tree/main/67-when-fair-funding-isnt

How do you fairly split the bill for something everyone benefits from?

- Picture an HOA board deciding whether to fund a new roof, a neighborhood group crowdfunding a footbridge, or a DAO voting to spend from a shared treasury.
- The standard textbook rule (the “pivotal mechanism”) says: build the project if pledges cover the cost, and charge each person only the smallest amount that was actually necessary for their vote to matter.
- This rule is provably honest for individuals — nobody benefits from lying about what they value the project at, acting alone.
- This paper asks the next question nobody had answered with an exact formula: what happens when a small group coordinates their pledges together?

Groups can game the textbook rule — this paper proves exactly when

- The “fair” critical-payment rule is not fair against coordinated groups: in the tested cases, coalitions of just two or three people broke it in the large majority of configurations.
- This paper proves — not just tests — exactly when that happens: whenever the project's cost is low enough that any one member's pledge is dispensable, the whole group can jointly game the price to zero.
- A simple, checkable formula tells a designer, before launch, whether their specific group size, pledge cap, and project cost are exposed — and exactly how much a colluding group stands to gain.
- Every claim is backed by an executable, independently re-checked proof — not a simulation and not an opinion.

Three neighbors, a \$3,000 footbridge

The rule's one loophole: you only pay if the bridge needed YOUR pledge specifically. Make sure it doesn't — even by lying — and you pay \$0.

IF EVERYONE IS HONEST

Alice \$0 + Bob \$1,000 + Carol \$1,000 = \$2,000
 $\$2,000 < \$3,000$ cost → nothing gets built, nobody pays.

IF ALL THREE LIE TOGETHER

Alice \$1,500 + Bob \$1,500 + Carol \$1,500 = \$4,500
 $\$4,500 \geq \$3,000$ cost → the bridge gets built.

SO WHAT DOES EACH OF THEM PAY?

Check Alice: cover her ears — Bob's \$1,500 + Carol's \$1,500 already = \$3,000. The bridge still gets built without her. So Alice wasn't needed → Alice pays \$0. The same is true for Bob and for Carol: any two of the three already clear \$3,000 alone. Nobody was ever individually needed, so nobody pays anything.

- Result: a real \$3,000 bridge gets built, and the group hands over \$0 total.
- This is not a bug in one bad rule — it is the standard, textbook “pivotal mechanism” taught in every mechanism-design course.

Exhaustive, machine-checked proof — not a sampled experiment

- **Exact enumeration.** Every possible allocation rule in the studied finite class is checked — all of them, not a sample — against six formal fairness and honesty properties.
- **A proven theorem.** A closed-form rule (not just a table of results) predicts exactly which allocation rules survive, for any group size and any pledge cap, and is confirmed against a larger untouched test domain.
- **Independent re-verification.** A second, separately written program re-derives every result from the raw data without reusing any of the first program's code — catching the kind of error a single implementation could hide.
- **An exact formula for group manipulation.** A closed-form expression for exactly how much a coalition of a given size can gain, derived by mathematical proof (convexity), not search — and it matches 145 independently brute-force-checked cases exactly.
- **A designer's checklist.** The paper ends with a plain checklist: plug in your group size, pledge cap, and cost, and know before launch whether your specific setup is exposed.

What this paper is, and deliberately is not

IS

- An exact, fully-enumerated finite theorem
- A reproducible, independently re-checked certificate
- A precise formula for coalition risk on this specific rule
- A practical pre-launch checklist for real designers

IS NOT

- A new universal impossibility theorem
- A claim about continuous (non-integer) values
- A repaired, collusion-proof mechanism
- Evidence about any specific deployed platform

Real systems already use this exact rule

- Participatory-budgeting platforms deciding which community proposal to fund.
- HOA and condo boards voting on a shared capital assessment.
- Crowdfunded neighborhood infrastructure — playgrounds, footbridges, shared solar installs.
- DAO treasuries and quadratic-funding platforms that have piloted pivotal-style payment rules.
- Anywhere a small, well-acquainted group decides together whether to build something and splits the bill by “what you were pivotal for” — this is the exact rule in question, and the paper's formula tells you if your setup is exposed.

Methodology and rigor

- **Certificate-first discipline.** Every accepted or rejected rule ships with a machine-checkable witness; nothing is asserted from sampled performance.
- **Independent replay.** A second implementation (`public_project_independent.py`) reconstructs every table and payment from serialized JSON without importing the primary verifier — zero mismatches across 145+ cross-checked rows.
- **SMT cross-checks.** For the neural-network/ReLU audit lane, an external exact-real Z3 solver independently confirms all reported extrema — 18 strict-bound counterexample queries, all unsat.
- **Preregistration.** The finite-lattice theorem and its confirmation domain were committed before the larger $n=3$, $m=4$ run; the coalition and false-name supplements are explicitly labeled post-hoc.
- **Full reproducibility.** Every script, config, and SHA-256 digest is published; the entire study reruns from a single command list.

[READ THE FULL PAPER](#)

SSRN Abstract ID 7293498

papers.ssrn.com/sol3/papers.cfm?abstract_id=7293498

Full code, data, and independent verification scripts:

github.com/jammysunshine/research-showcase/tree/main/67-when-fair-funding-isnt

Mohit Mendiratta — Independent research